

Exercise 1: We call the Gamma function the function defined by

$$\Gamma : x \mapsto \int_0^{+\infty} e^{-t} t^{x-1} dt.$$

1. Show that Γ is defined on $]0, +\infty[$.
2. Show that Γ is continuous on $]0, +\infty[$.
3. Show that Γ is of class C^∞ on $]0, +\infty[$ with $\forall x \in N^*, \forall x \in R_+^*, \Gamma^{(k)}(x) = \int_0^{+\infty} (\ln t)^k e^{-t} t^{x-1} dt$.
4. Show that for all $x > 0, \Gamma(x+1) = x\Gamma(x)$. Deduce that $\forall n \in N^*, \Gamma(n) = (n-1)!$.
5. Calculate $\Gamma(\frac{1}{2})$.

Exercise 2:

1. Show that

$$\int_0^1 (\ln(\frac{1}{y}))^{n-1} dy = \Gamma(n), \quad \int_0^{+\infty} x^n e^{-k^2 x^2} dx = \frac{1}{2k^{n+1}} \Gamma(n).$$

2. Calculate the following integrals:

$$\int_0^{+\infty} e^{-x^2} dx, \quad \int_0^{+\infty} \sqrt{x} e^{-3\sqrt{x}} dx.$$

Exercise 3: Calculate the following integrals

$$\begin{aligned} \int_0^{+\infty} x^5 e^{-x^4} dx, & \quad \int_0^{+\infty} x^3 e^{-\frac{x^2}{2}} dx, & \quad \int_0^{+\infty} 2^{-3x^2} dx, & \quad \int_0^{+\infty} \sqrt{2} e^{-\sqrt{x}} dx, \\ \int_1^{+\infty} \frac{(\ln x)^3}{x^2} dx, & \quad \int_0^{+\infty} 2\sqrt{x} e^{-x^2} dx, & \quad \int_0^{+\infty} x^6 e^{-4x^2} dx, & \quad \int_0^1 \left(\ln \frac{1}{x}\right)^{a-1} dx \end{aligned}$$

Exercise 4: Calculate the following integrals

$$\begin{aligned} \int_0^{\pi/2} \sin^2 \theta d\theta, & \quad \int_0^{\pi/4} \sin^3(2x) \cos^4(2x) dx, & \quad \int_0^{\pi} \sin^5\left(\frac{x}{2}\right) \cos^2\left(\frac{x}{2}\right) dx, & \quad \int_0^4 \frac{x^3}{\sqrt{4-x}} dx, \\ \int_0^1 \frac{4}{\sqrt[4]{1-x^4}} dx, & \quad \int_0^1 x^3 (1-\sqrt{x})^5 dx, & \quad \int_0^1 \frac{x}{\sqrt{1-x^3}} dx, & \quad \int_0^1 (x \log x)^3 dx, \\ & \quad \int_3^7 \sqrt[4]{(x-3)(7-x)} dx, & \quad \int_0^e \frac{x}{\sqrt{1-\ln x}} dx \end{aligned}$$